

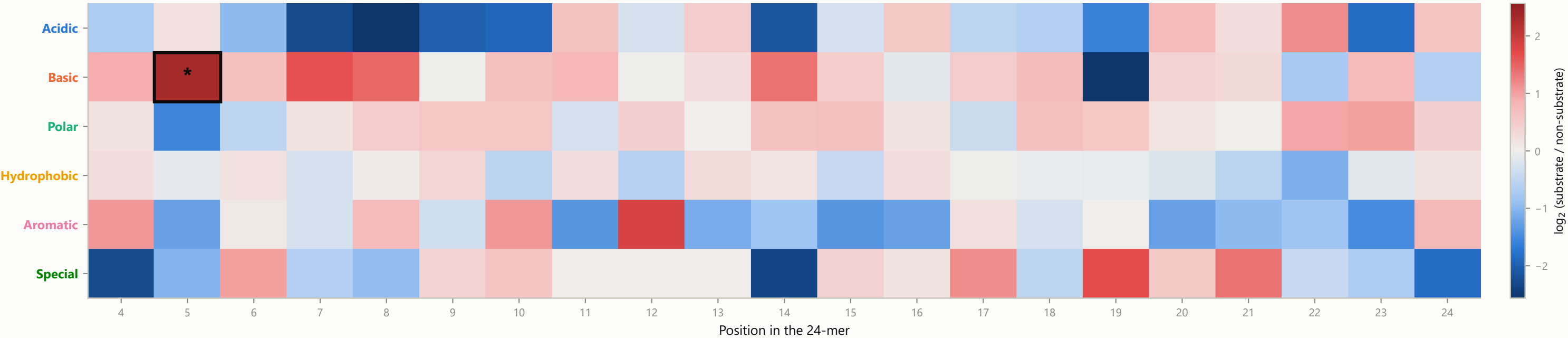
Positions 4-24: what separates motif-bearing substrates from motif-bearing non-substrates

Both groups carry M-[P/G]-[E/D], so positions 1-3 are identical by construction and only the downstream window can differ. No single residue or position separates them, but the substrates carry a markedly more positive net charge across the window.

A

Chemical class by position, downstream of the motif

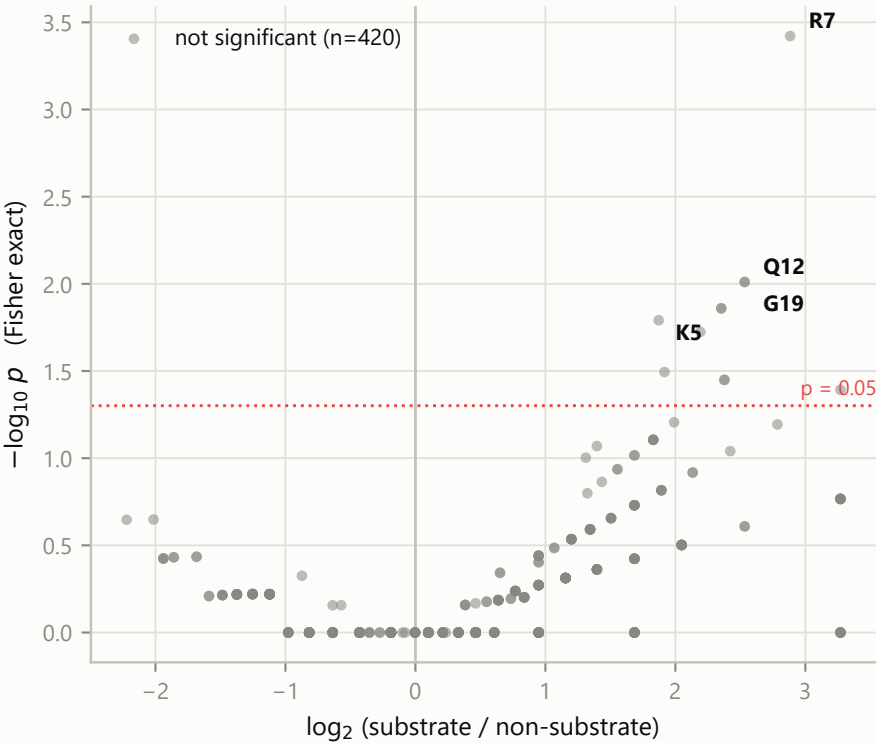
16 motif-bearing substrates vs 163 motif-bearing non-substrates. Boxed * = $q < 0.05$ (1 of 126 cells): Basic at position 5.



B

No individual residue distinguishes them

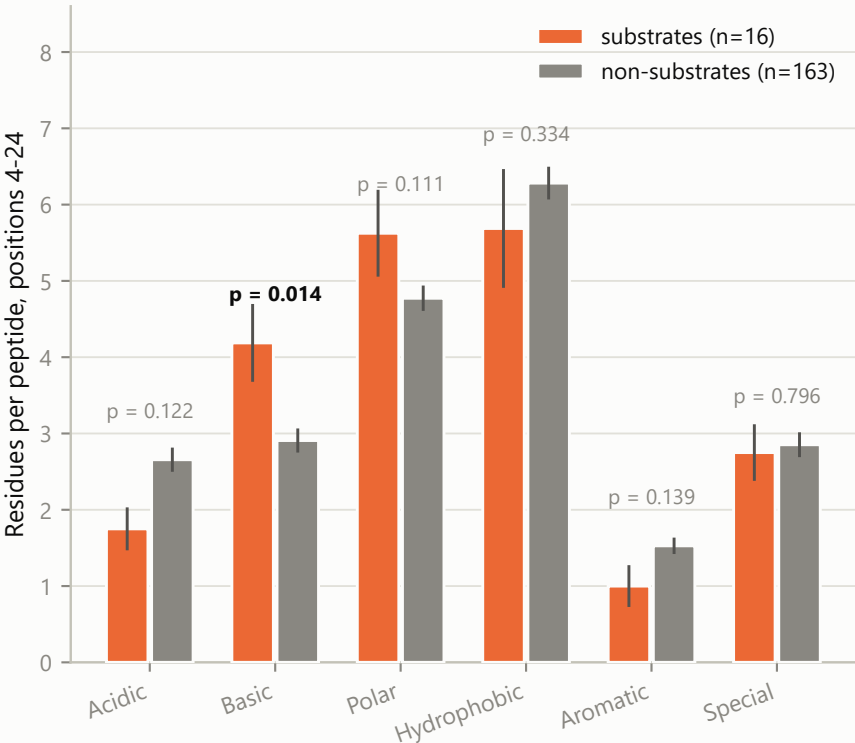
11 of 420 cells reach $p < 0.05$ - fewer than the 21 expected by chance; none survive FDR



C

Only basic residues differ, and only in aggregate

Mean per peptide over positions 4-24; bars are SEM, p from Mann-Whitney



D

Substrates carry a net positive charge downstream

Both groups carry the motif - only the downstream window differs

